

Male Reproductive System

Human testis: (Slide # 8-9 H&E + diagram)

- The outer surface of testis is lined by a thin squamous mesothelium (tunica vaginalis that covers the anterior and lateral aspects of testis), beneath which is a very dense irregular capsule called the **tunica albuginea**. Connective tissue septa subdivide the interior into about 250 lobular compartments. Centrally the septa merge forming the vascular and nerve rich mediastinum (mediastinum testis).
- Each lobular compartment contains 2-4 coiled, compacted seminiferous tubules. Each seminiferous tubule is approximately 50 cm long and 200 μ m. The seminiferous tubules consist of a seminiferous epithelium surrounded by a tunica propria of myoid cells. The seminiferous epithelium is a complex stratified epithelium composed of Sertoli cells and spermatogenic cells. The large pyramidal Sertoli cells can be identified by the large elliptical nucleus with prominent nucleolus. The testosterone secreting interstitial cells (Leydig cells) as well as the blood vessels, nerves and lymphatics are in the connective tissue external to the tunica propria.
- Spermatogonia can be identified in the basal compartment of the seminiferous tubule. Type A dark spermatogonia have ovoid nuclei with intense basophilic, granular chromatin. Type A pale spermatogonia have ovoid nuclei with lightly staining, granular chromatin. Type B spermatogonia have spherical nuclei with chromatin condensed into clumps. You are only required to identify spermatogonia in general, not the different types.
- Primary spermatocytes can be identified by the distinct appearance of the chromatin. These large cells undergo meiosis I and the chromatin is condensed into visible chromosomes.
- Secondary spermatocytes undergo meiosis II to produce haploid spermatids. Secondary spermatocytes are found closer to the lumen, are about half the size of the primary spermatocytes and are frequently not identifiable because they rapidly divide to form the spermatids.
- Early spermatids are small rounded cells near the lumen with round nuclei. These cells undergo remodeling as they differentiate into late spermatids and spermatozoa. During spermiogenesis the late spermatid nuclei condense and become more pointed. The heads of these forming spermatozoa are embedded in the Sertoli cell cytoplasm and their flagella extend into the lumen.

Mediastinum: (Slide #10) H&E

- This region receives the distal portions of seminiferous tubules. At the end of each seminiferous tubule there is an abrupt transition to the straight tubules (**tubuli recti**). This short section is lined only by Sertoli cells. The straight tubules empty into the rete testis. As they near the rete testis, the epithelium changes to simple cuboidal epithelium.

Rete Testis: (Slide #10) H&E

- Refers to complex series of interconnecting channels that are enclosed within the highly vascular connective tissue of the mediastinum. These channels, which can be visualized as slit like areas are lined by simple cuboidal or low columnar epithelium. The epithelial cells lining these channels possess a single apical cilium and relatively few short apical microvilli.

Identify and check-off each of the following: Tunica albuginea, tunica vasculosa, septa, lobule, tunica propria, seminiferous tubule, interstitium with blood vessels, Leydig cells, Sertoli cells, spermatogonia, primary spermatocytes, early spermatids, late spermatids, spermatozoa

Efferent Ductules: (Slide #10) H&E + picric acid, trichrome

- There are approximately 12-20 efferent ductules that connect the channels of the rete testis to the superior pole of the mediastinum to form the proximal portion of the epididymis.
- The efferent ductules are lined with a pseudostratified columnar epithelium that contain clumps of tall and short cells, giving the luminal surface a sawtooth appearance.
- Interspersed among the columnar cells are occasional basal cells that serve as epithelial stem cells.
- The tall columnar cells are **ciliated**. The short nonciliated cells have numerous microvilli. Most of the fluid secreted in the seminiferous tubules is reabsorbed in the efferent ductules.
- A smooth muscle layer first appears at the beginning of the efferent ductules. This soon becomes several layers to form a circular sheath in the wall of the ductules. Interspersed among the muscle cells are elastic fibers.
- Ciliary action and contraction of the fibromuscular layer enable the transport of sperm along the duct system.

Epididymis: Slide # 11 (H&E)

- Epididymis is divided as head, body and tail regions. The head is practically formed by all the efferent ductules converging to form the duct of the epididymis. The epididymis is 4-6 meters long tubule that is highly coiled. Due to this coiled nature, several profiles of the duct can be visualized in the slide, under low magnification.

- This duct is lined by pseudostratified columnar epithelium. There are two types of cells found in this epithelium. They are the basal cells and the columnar cells (principal) with **stereocilia**.
- The lining columnar cells possess numerous stereocilia projecting from their apical surfaces. They help in reabsorption of excess testicular fluid and helps secrete glycerophosphocholine, a substance that promotes sperm maturation.
- The basal cells serve as stem cell population.
- The junctional complexes exhibited by the apical surfaces of the columnar cells, form a uniform border towards the luminal surface.
- Spermatozoa are normally seen in the lumen in most sections.
- The epithelium is contained within a sleeve of smooth muscle. The muscle is thinnest at the proximal end of the epididymis and progressively thickens close to the vas deferens.
- In addition, migrating lymphocytes called “halo cells” are often found within the epithelium.

Identify and check-off each of the following: Efferent ductules, epididymis, pseudostratified epithelium with cilia, pseudostratified epithelium with stereocilia

Spermatic Cord with ductus deferens: Slide # 11 (H&E)

The spermatic cord consists of the ductus deferens, blood vessels, nerves, the pampiniform plexus (temperature regulation) and cremaster muscle fibers.

The **ductus deferens** is lined by a pseudostratified columnar epithelium (columnar cells with stereocilia), which lines a thick-walled muscular tube.

The muscles are organized as inner and outer longitudinal layers and a middle circular layer with respect to the long axis of the duct. Muscular contractions of this duct account for most of the ejaculatory pressure.

Identify the plexus of large irregular veins. The pampiniform plexus is a counter current exchange-cooling device to lower blood temperature. The veins have a thick muscular wall with two layers: an inner longitudinal layer and outer circular layer. Bundles of peripheral nerves and adipose are found within the loose CT of the cord. Striated skeletal muscle of the cremaster muscle may also be observed.

The distal portion of ductus deferens is dilated to form the ampulla. This portion is characterized by tall, branched mucosal folds that often show glandular diverticula.

Identify and check-off each of the following: Spermatic cord, ductus deferens, pseudostratified columnar epithelium with stereocilia, pampiniform plexus.

Seminal vesicle: Slide # 12 (H&E)

Seminal vesicle is an accessory sex gland. They are paired. The duct of each seminal vesicle joins the ampulla of the vas deferens to form ejaculatory duct, which opens into prostatic urethra.

Seminal vesicle is an elongated, highly folded tubular structure of about 15-20 cm. Due to the convoluted nature, several profiles of this tubular structure can be visualized in low magnification.

The wall of seminal vesicle contains a mucosa, a thin layer of muscle and a fibrous coat.

The mucosa is thrown into numerous primary, secondary and tertiary folds that increase the secretory surface area. All the irregular chambers thus formed, however, communicate with the lumen.

The epithelium consists of tall nonciliated columnar cells and short, round cells that rest on basal lamina. Columnar cells are secretory, and the short ones are the stem cells. The cells secrete about 70% of the fructose-rich semen.

The epithelial cells possess lipochrome pigments which give the secretion a yellow color. The secretory function and morphology of the seminal vesicles are under the control of testosterone.

Identify and check-off each of the following: Seminal vesicle, pseudostratified columnar epithelium, smooth muscle layers.

Prostate: Slide #14 (H&E)

The prostate, the largest accessory sex gland in the male, surrounds the urethra from its origin from the bladder. The gland consists of 30-50 tubuloalveolar glands drained by 20 or more ducts into the prostatic urethra.

The gland is enclosed in a CT capsule subdivided by septa. The glands are irregularly organized, and the lining epithelium is variable with cuboidal, columnar and pseudostratified epithelium observed.

The alveoli of the prostatic glands, especially those in older men, often contain concentric lamellated bodies - prostatic concretions (**corpora amylacea**) of varied shape and size.

The lamina propria surrounding the glands is rich in smooth muscle cells. The prostatic urethra is lined by transitional epithelium containing an eosinophilic apical cell layer.

The glands are arranged in three strata surrounding the urethra (beneath the epithelium are periurethral or mucosal glands, around which are the transitional or submucosal glands and to the peripherally are the more abundant peripheral or main prostatic glands.

The glandular epithelium is influenced by sex hormones such as testosterone and adrenal androgens.

Identify and check-off each of the following: Prostate gland, glandular epithelium, prostatic concretions, fibromuscular stroma